DESIGN SYSTEMS

TECHNICAL DATA SHEET TIX-CEM
CEMENTITIOUS EPOXY-POLYAMIDE DECORATIVE SYSTEM

DESCRIPTION

Three-component, water-based cementitious epoxy-polyamide decorative system.

GENERAL CHARACTERISTICS

Three-component product formulated with modified epoxy resins in emulsion, hydraulic binders, inert silicate salts and a polyamide hardener. It has high bonding power, good water resistance and excellent adhesion on new and old concrete surfaces. TIX-CEM is a product formulated to adhere perfectly to old concrete and also to new, not fully cured concrete, tiles, bricks and stone. It is an excellent surface treatment, promoting the bonding of protective, anti-corrosion and waterproofing coatings. It has excellent waterproofing capacity and forms a membrane system that allows the migration (permeability) of vapour while at the same time resisting positive and negative H2O pressure.

FIELDS OF APPLICATION

TIX-CEM is specifically for the preparation of concrete substrates for the construction of epoxy or polyurethane resin floors. The absence of solvents makes it suitable for substrate preparation, even in enclosed spaces, such as residential and industrial floors, walls and floors of cold rooms, concrete tanks and food storage facilities. Excellent for preparing brick or concrete surfaces to create multi-layer finishes with graded sand.

Mixed with spheroidal quartz of controlled particle size, it creates the TIX-CEM system for the construction of coatings with a pleasing design impact and a characteristic texture.

PREPARATION

The preparation of the surface to be treated is of fundamental importance. TIX-CEM must be applied on dry or slightly damp substrates, well cleaned and free from oils, grease, paint residues and other removable parts. If necessary, remove every trace of any contaminant. The substrate must have been laid at least 15 days earlier. Poor preparation compromises the adhesion of the product.

PHYSICAL PROPERTIES (TYPICAL VALUES)

CHARACTERISTIC	VALUE
Mixing ratio by weight A : B (A = base; B = hardener)	2 : 1 (by weight)
Pot life at 20 °C	40 - 60 min
Adhesion to concrete (UNI EN 1542-1766)	> 3 N/mm^2 (concrete failure)
Compressive strength (after 28 days)	300 kg/cm^2
Vapour permeability (UNI EN ISO 7783-2)	μ > 7000

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Resistance to positive H2O pressure	7,5 kg/cm^2
Dust-free time at 25 °C	4 hours
Overcoating time at 25 °C	12 hours
Full curing time at 25 °C	6 days
Application temperature	from +5 °C to +35 °C
Thermal shock resistance (cured product)	from -15 °C to +90 °C
Flammability of the product	Non-flammable

APPLICATION

TIX-CEM can be applied by trowel without any dilution, or by brush, roller or spray depending on the specific requirements, and can be diluted from 5% to a maximum of 20% depending on the use, using drinking water.

Apply the product in ambient conditions with a temperature between +5 °C and +35 °C and relative humidity between 35% and 75%.

When opening each pack, stir the product thoroughly to remove any sediment.

Then mix the two components A (resin) and B (hardener) thoroughly in the indicated ratio. Next, add the third component C (powder) up to a maximum of 15 kg, then mix until a homogeneous, lump-free mix of uniform colour is obtained. After mixing the components, the product is ready for use. If the product is very viscous, reduce the quantity of component C.

The product can be diluted from 5% to 15% (on A+B) depending on the use, using drinking water.

The pot life of the catalysed product, i.e. the usable time for application, is approx. 40-60 minutes (in ambient conditions); depending on the application, it is advisable to prepare an adequate quantity of product so as to guarantee optimum workability.

Drying reaches the "dust-free" stage in 4 hours at 25 °C, it can be overcoated after 12 hours at 25 °C, full curing in 6 days at 25 °C.

IMPORTANT NOTE: the quality of the application will be influenced by the tools, equipment and instruments used, as well as by the ambient application conditions, such as temperature, relative humidity and the presence of dust in the application area. When applying the product for the first time, it is recommended to carry out some preliminary tests in order to check and validate the process, to guarantee the best technical and aesthetic result.

COVERAGE

Coverage depends on the type and degree of roughness of the substrate to be treated. On average, approx. 1,5 kg/m^2 (A+B+C) of product is required to obtain a thickness of 1 mm. For smooth surfaces, such as tiles and flat concrete, approx. 0,5-0,6 kg/m^2 is sufficient.

DESIGN SYSTEMS

PACKAGING

30 kg system consisting of pails:

- Component A: 10 kg;
- Component B: 5 kg;
- Component C: 15 kg.

STORAGE

Protect from frost, heat and direct sunlight. In original, intact containers, protected from frost and excessive heat, components A and B keep for approx. 12 months; component C for approx. 6 months.

PRECAUTIONS AND WARNINGS

Do not eat, drink or smoke during application. Use personal protective equipment: gloves, goggles, mask and clothing of the type indicated in the safety data sheet.

Always comply with the regulations and provisions in force on safety and transport. Consult the technical data sheet, the application guidelines and the safety data sheet. For any clarification or information, please contact our Technical Customer Service.

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